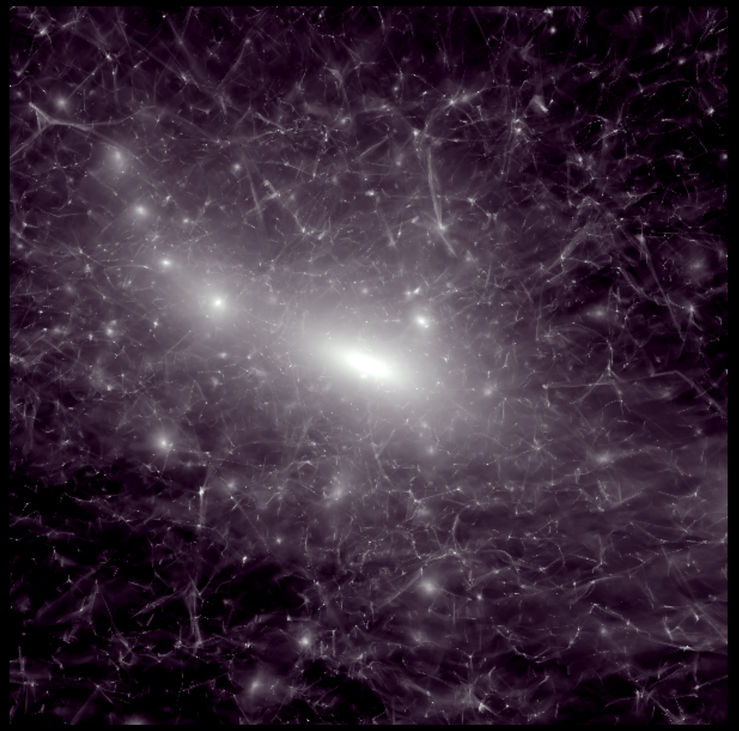
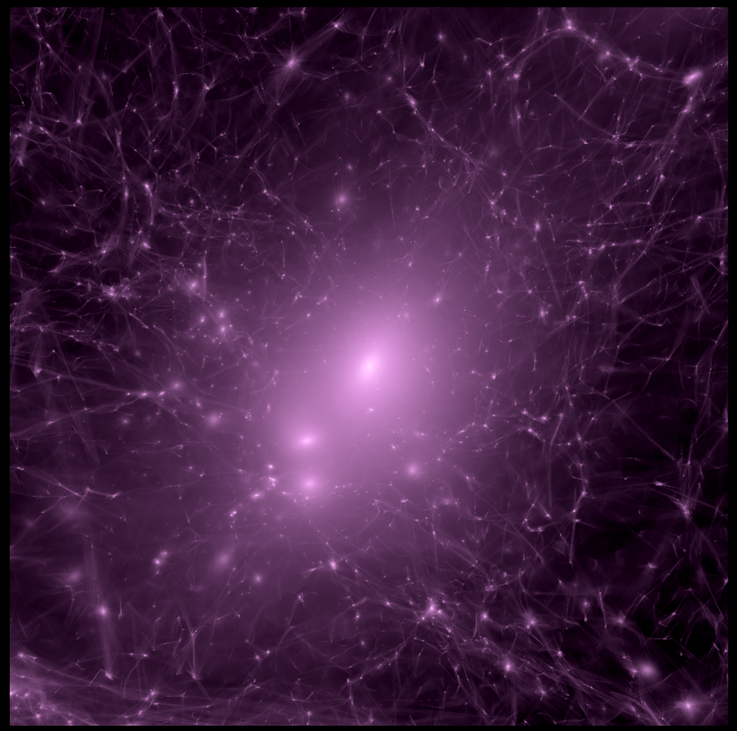
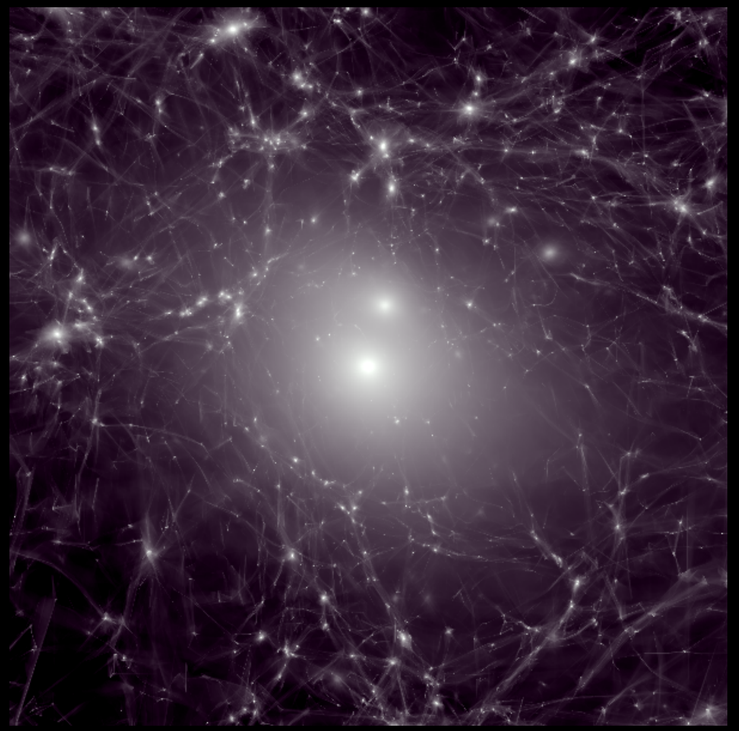
AstroReach SD: Dark Matter & Python





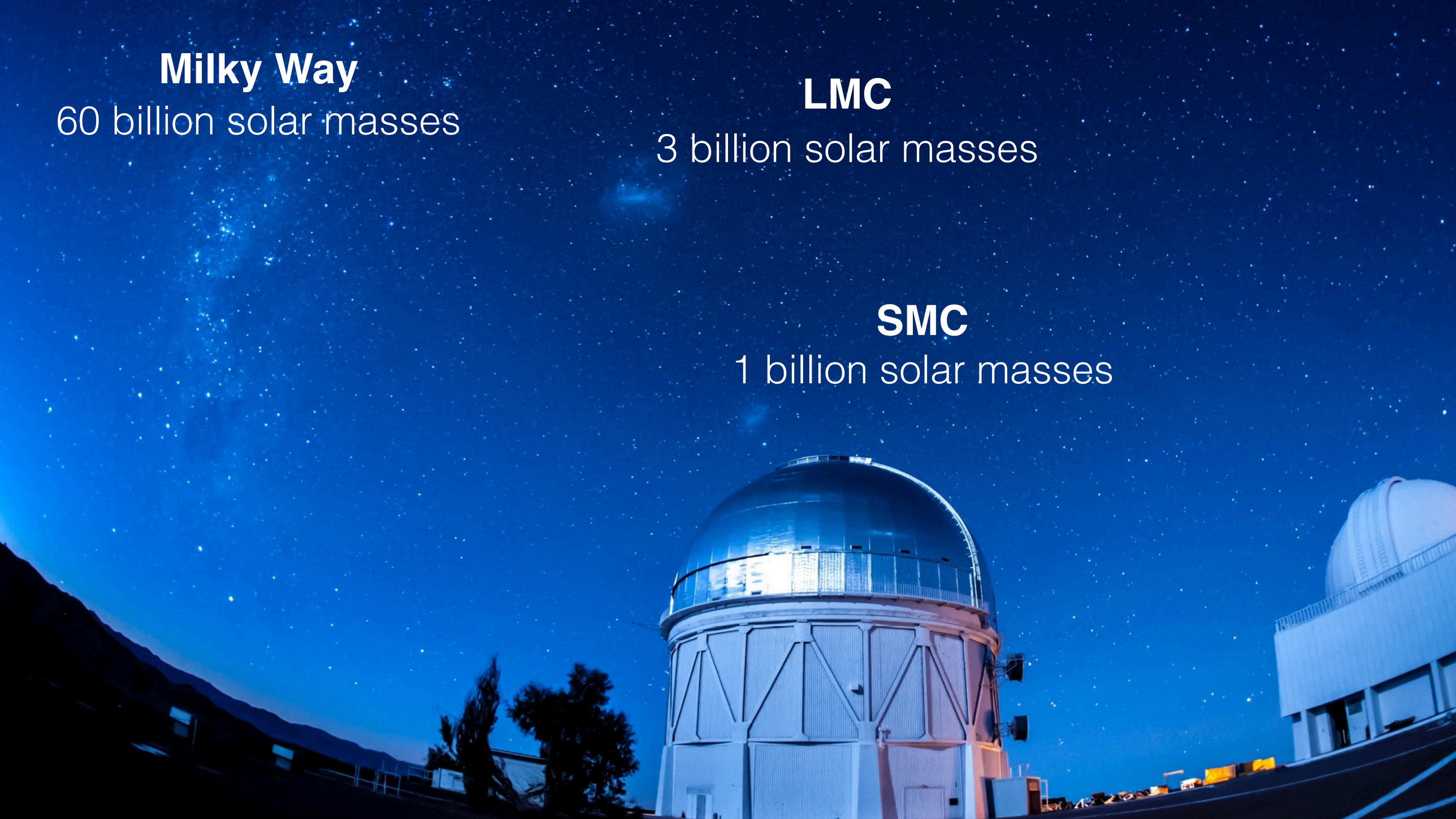


THE PREUSS SCHOOL

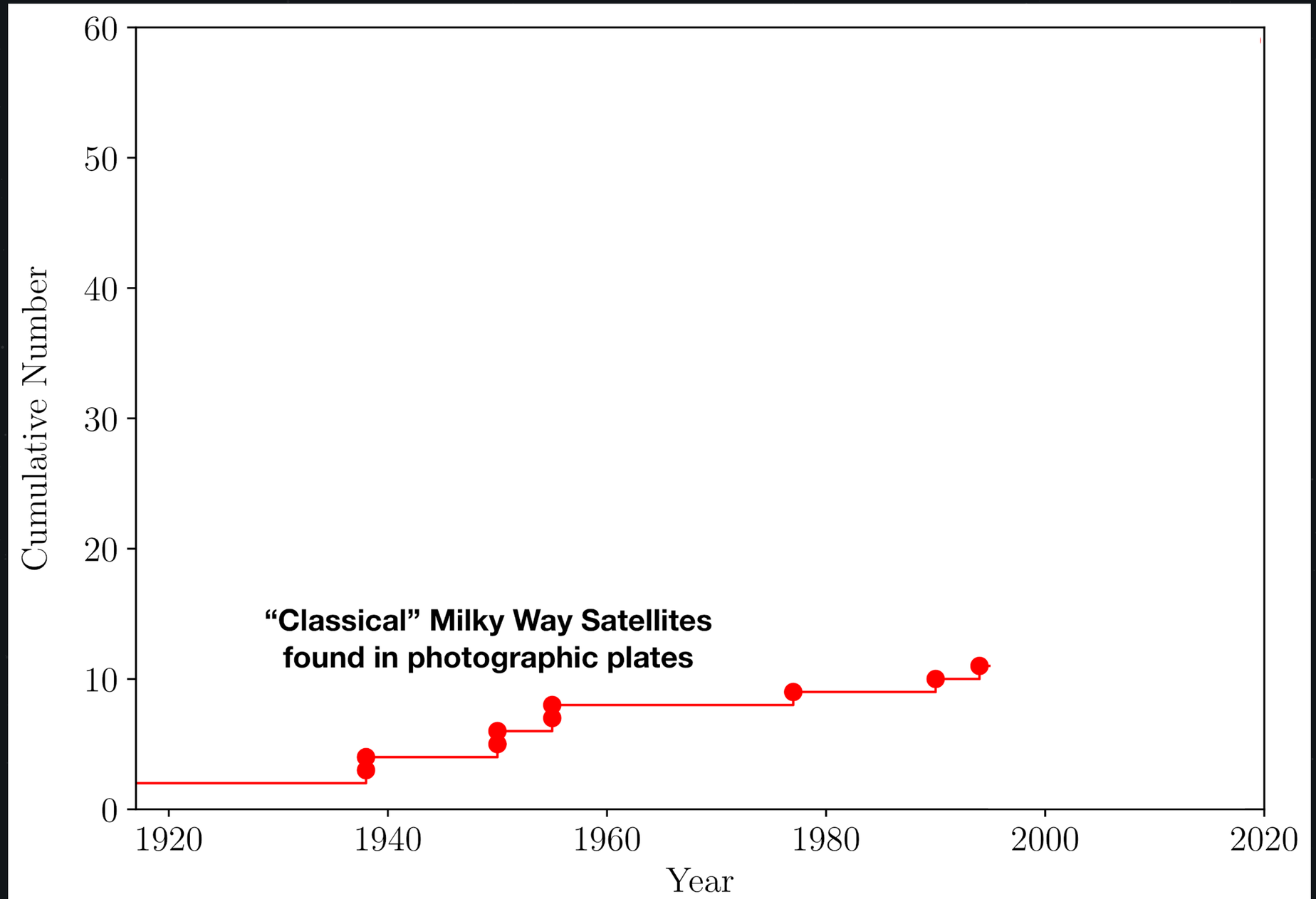
Week 5: Dwarf Galaxies & Dark Matter
02/11/2026

UC San Diego

Milky Way
60 billion solar masses


Milky Way
60 billion solar masses
LMC
3 billion solar masses
SMC
1 billion solar masses

Satellite Galaxies Orbiting the Milky Way



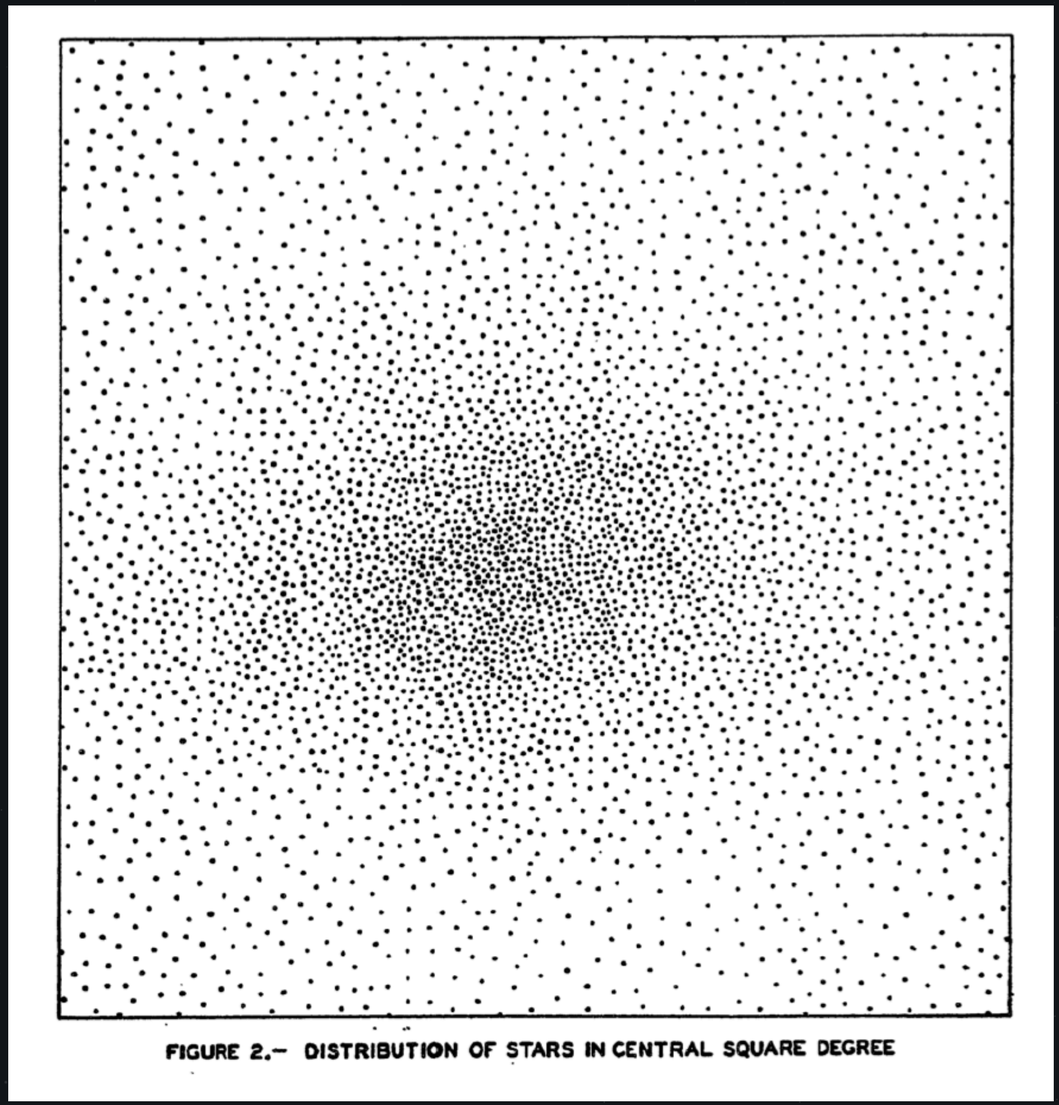


FIGURE 2.— DISTRIBUTION OF STARS IN CENTRAL SQUARE DEGREE

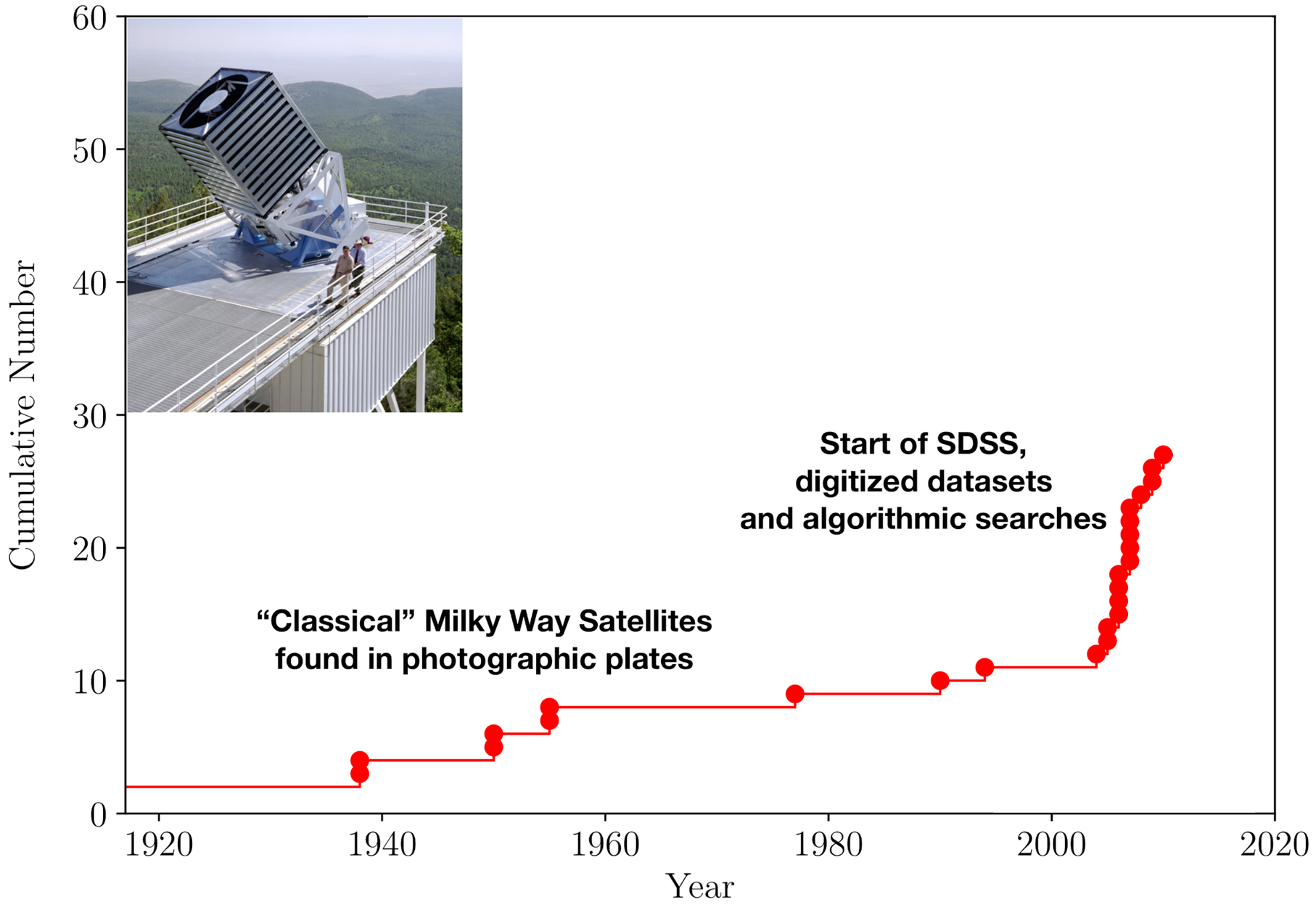
Sculptor

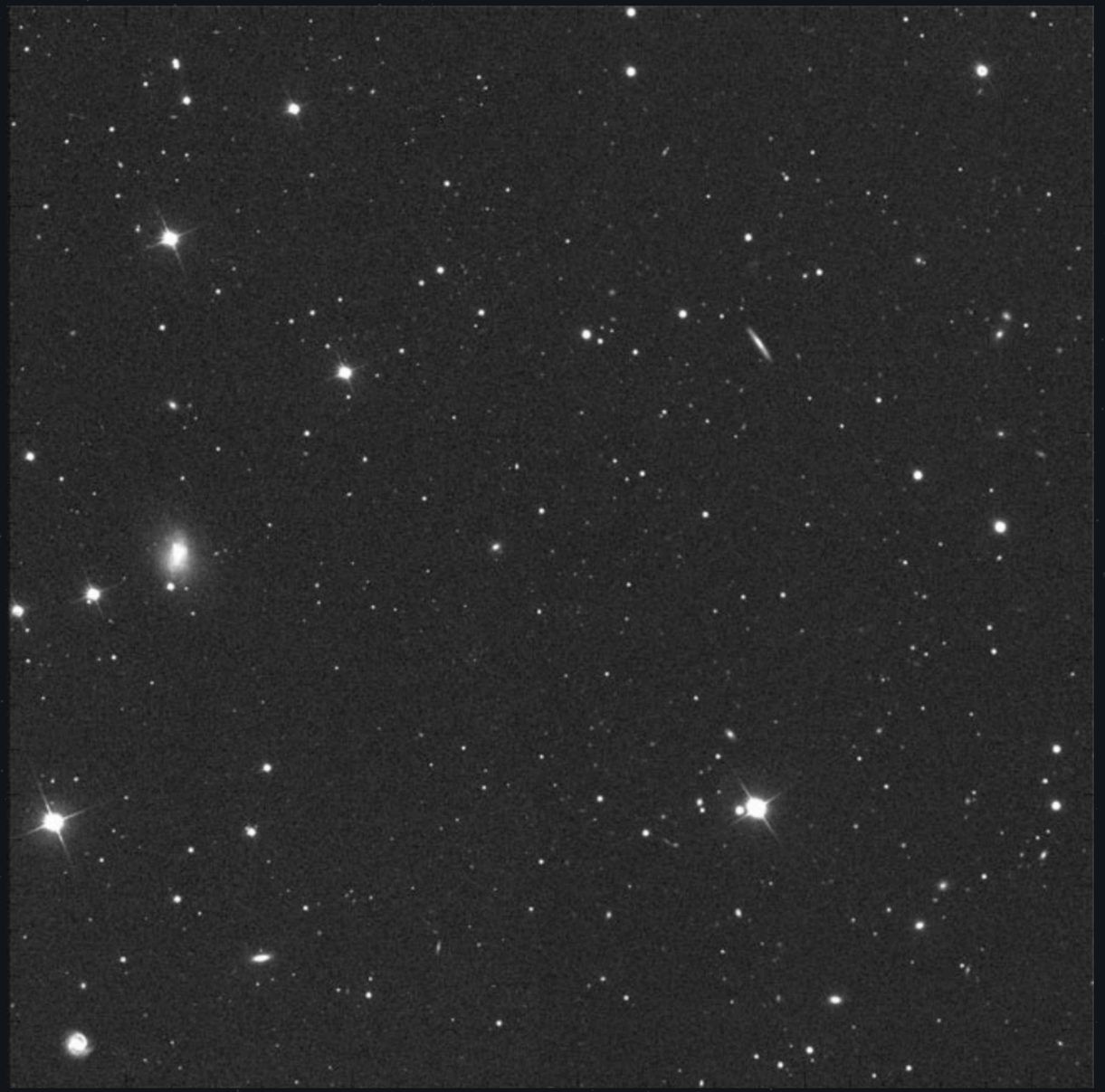
4 million suns

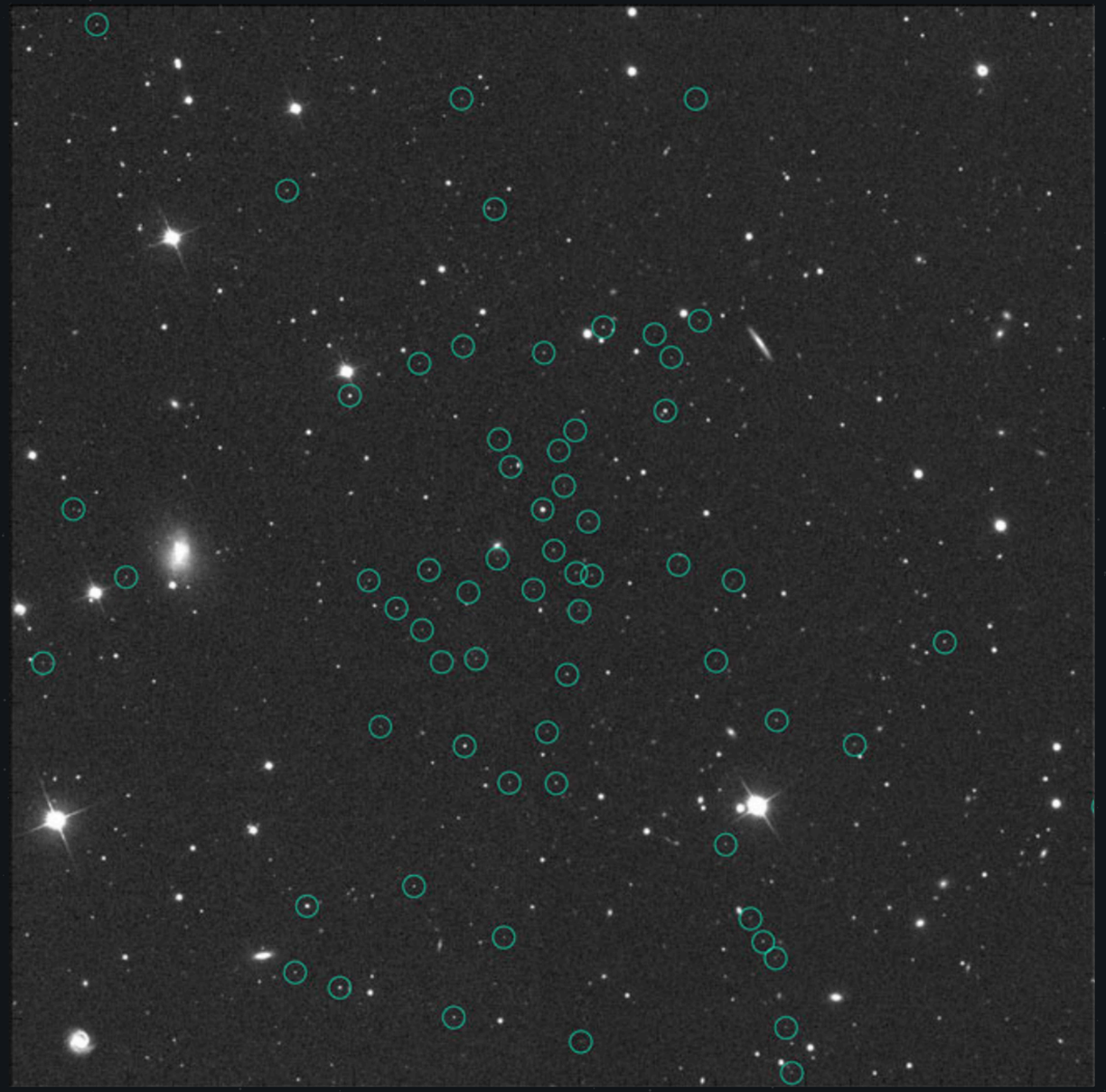
In many respects it appeared to be unlike any known stellar organization. The finding more recently of a similar system in Fornax ... suggests that a description of these objects may be of general interest.

- Harlow Shapley, 1938

Satellite Galaxies Orbiting the Milky Way







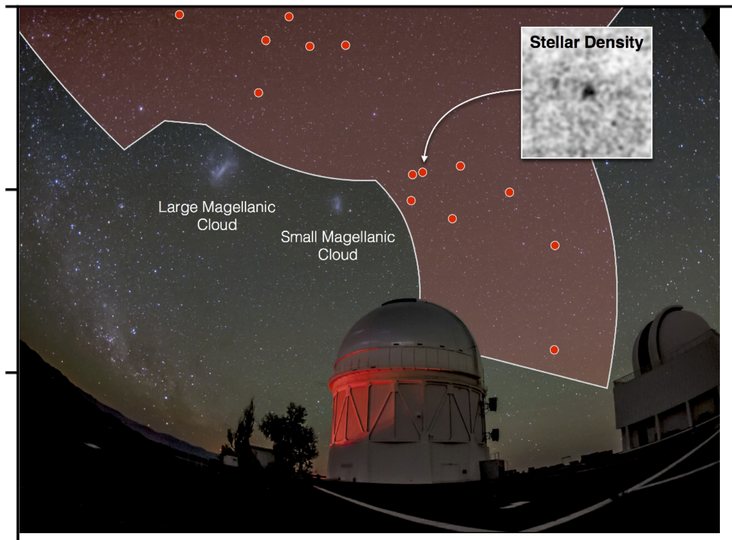
Segue I

300 suns

the darkest known galaxy!

THE DARK ENERGY SURVEY
2012-2018

Satellite Galaxies Orbiting the Milky Way



New generation of optical surveys with wider and deeper imaging: DES, SMASH, MagLiteS, ATLAS, Pan-STARRS, HSC, Gaia and others…

Start of SDSS, digitized datasets and algorithmic searches

“Classical” Milky Way Satellites found in photographic plates

Confirmed

Candidate

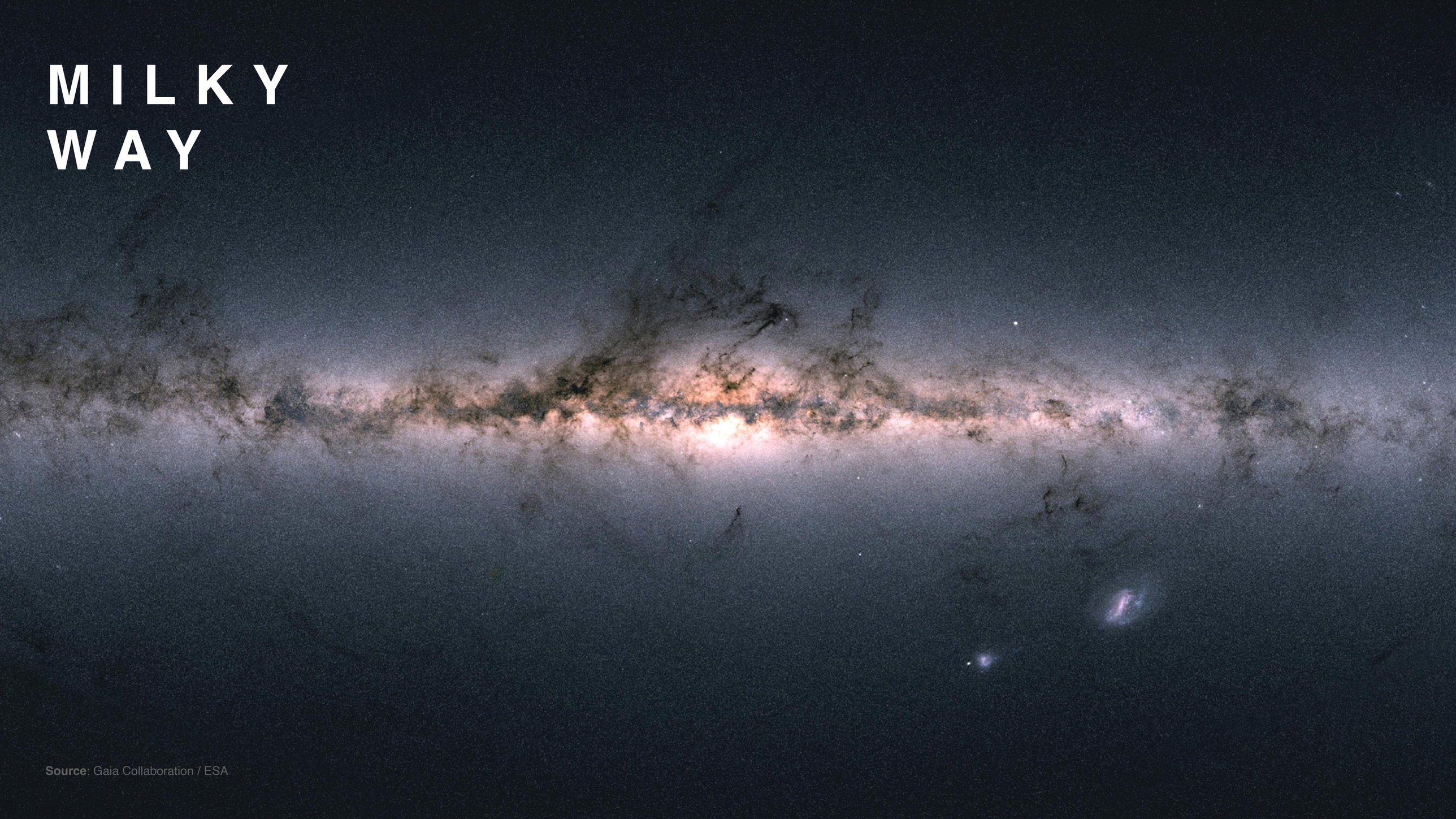
Cumulative Number

0 10 20 30 40 50 60

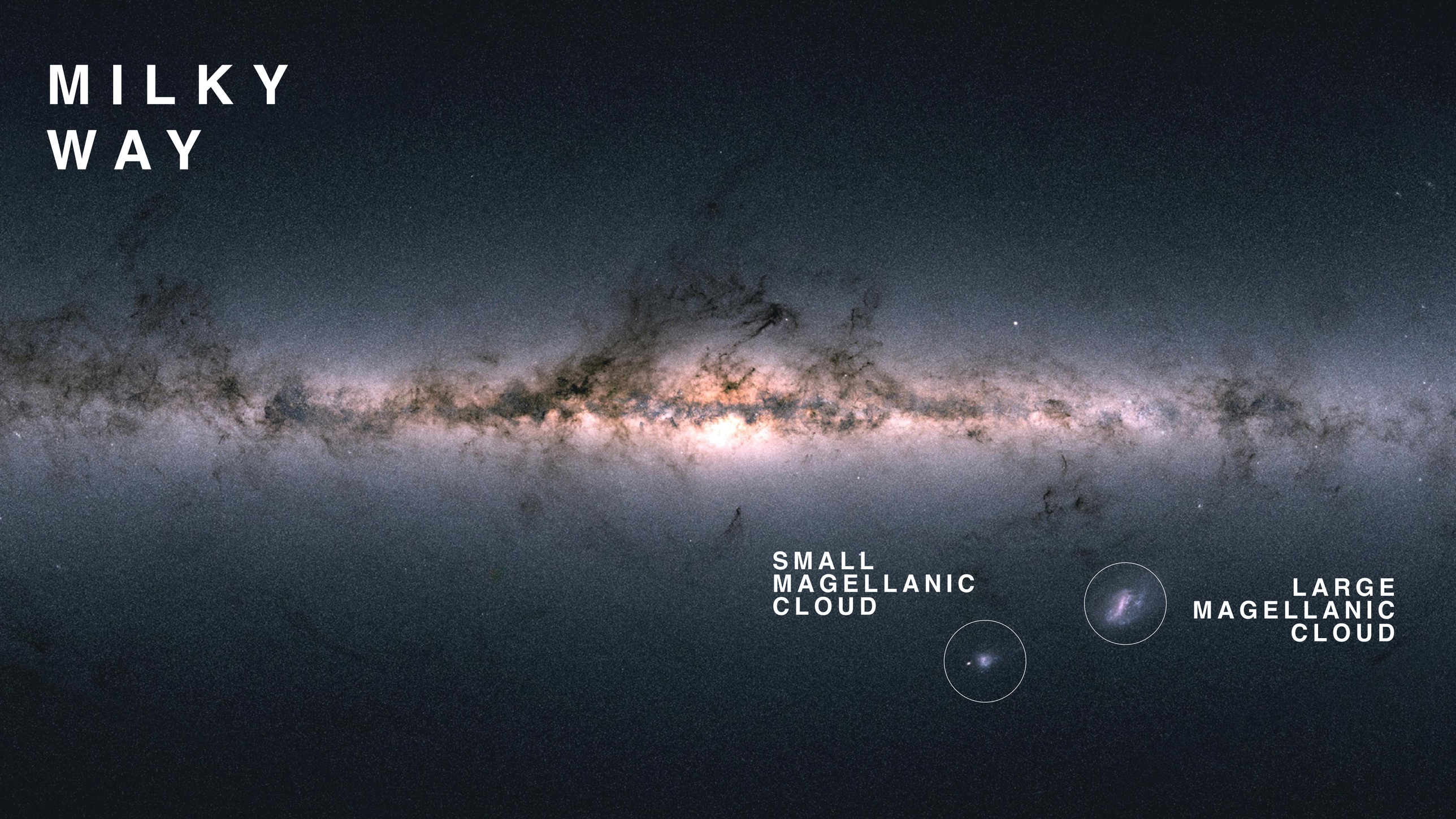
1920 1940 1960 1980 2000 2020

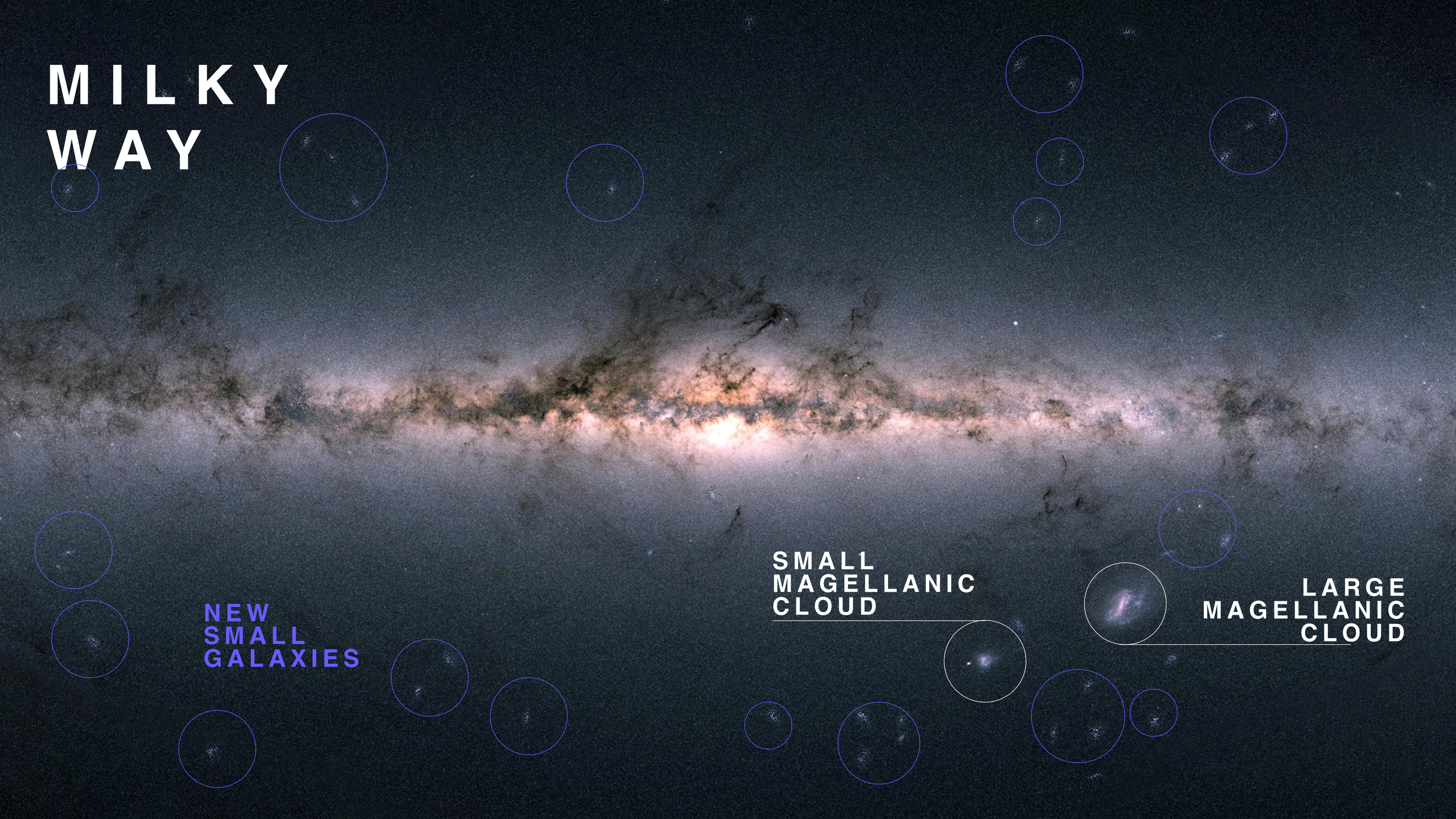
Year

MILKY WAY

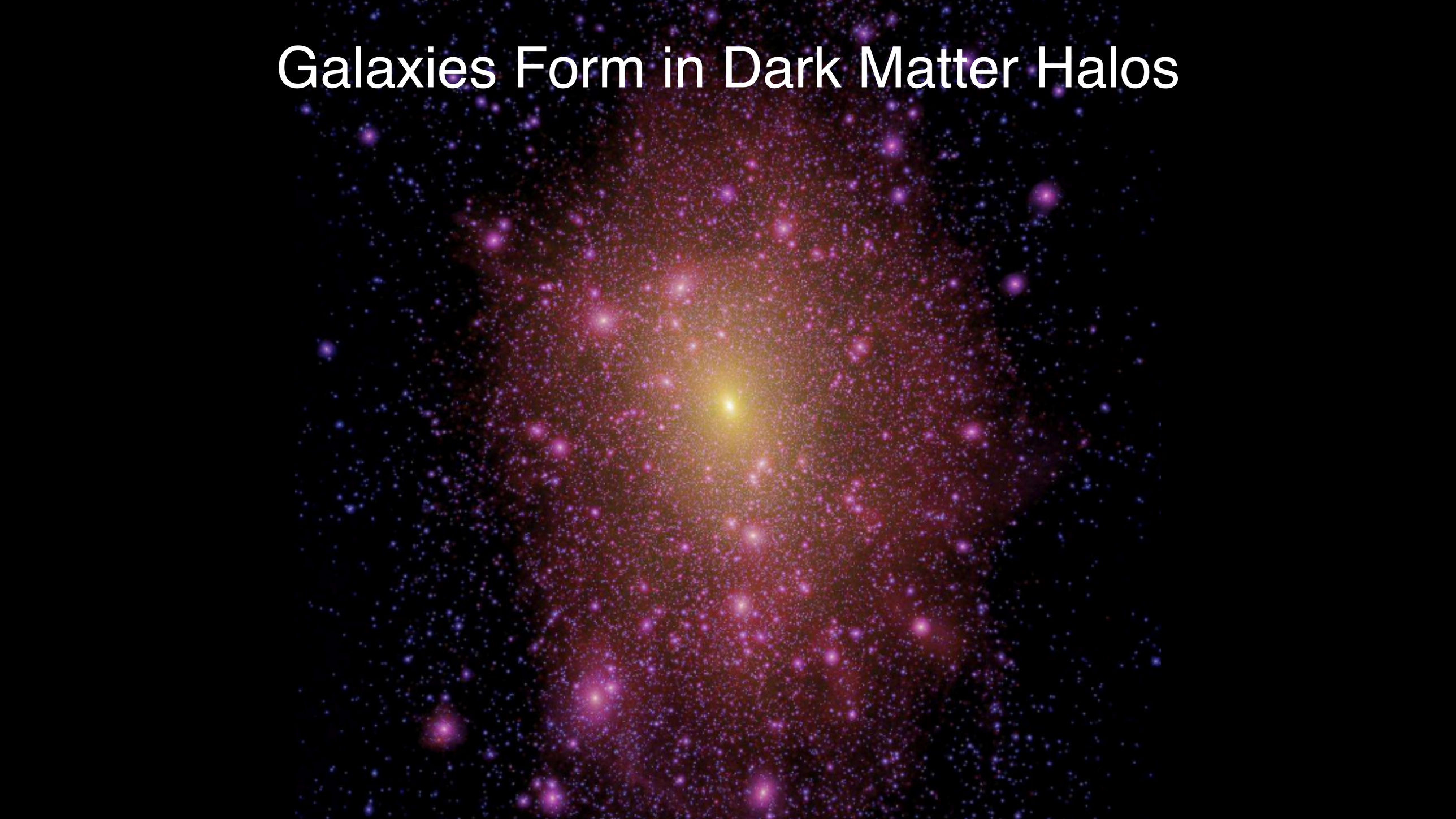


Source: Gaia Collaboration / ESA


MILKY
WAY
SMALL
MAGELLANIC
CLOUD
LARGE
MAGELLANIC
CLOUD


MILKY
WAY
NEW
SMALL
GALAXIES
SMALL
MAGELLANIC
CLOUD
LARGE
MAGELLANIC
CLOUD

Galaxies Form in Dark Matter Halos

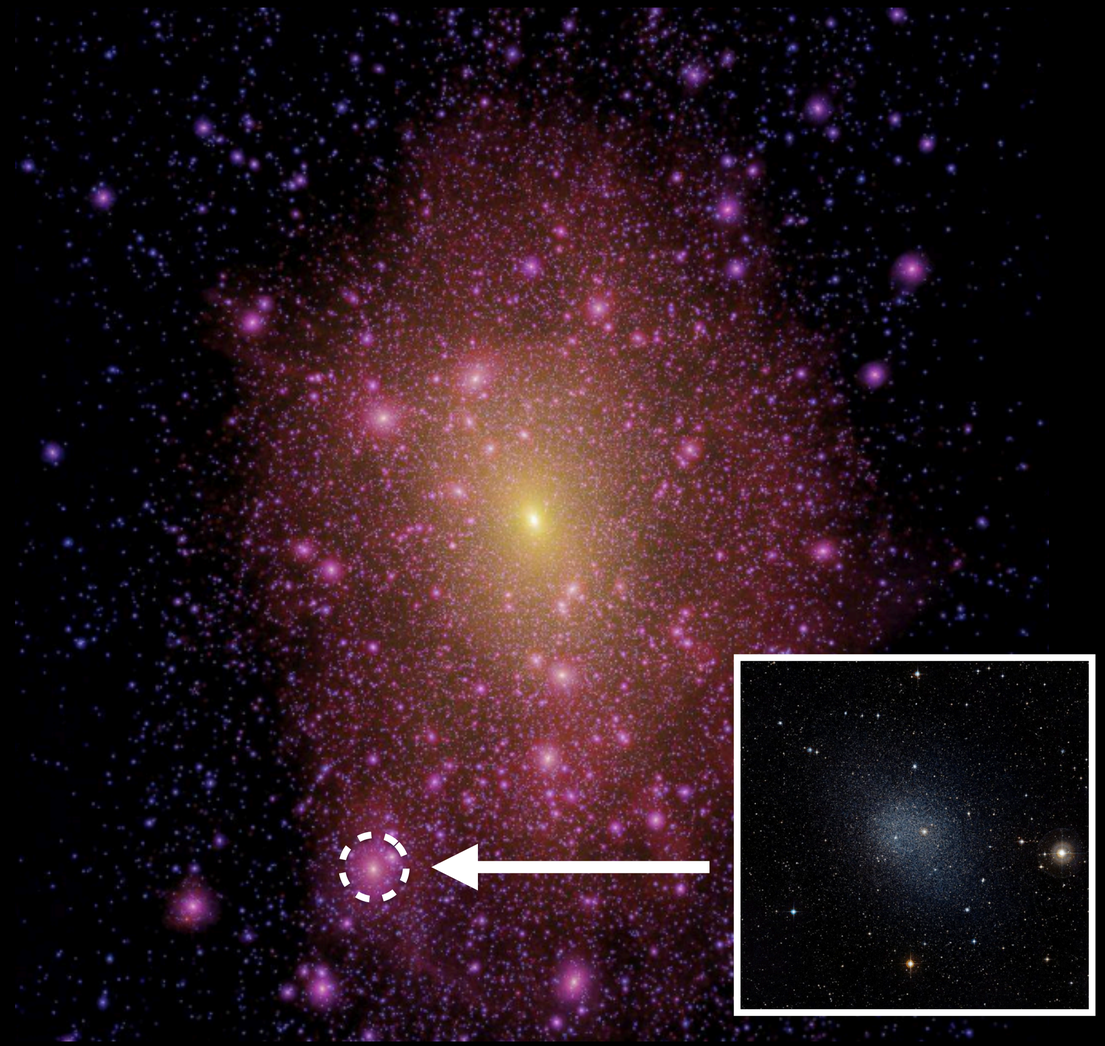


Galaxies Form in Dark Matter Halos

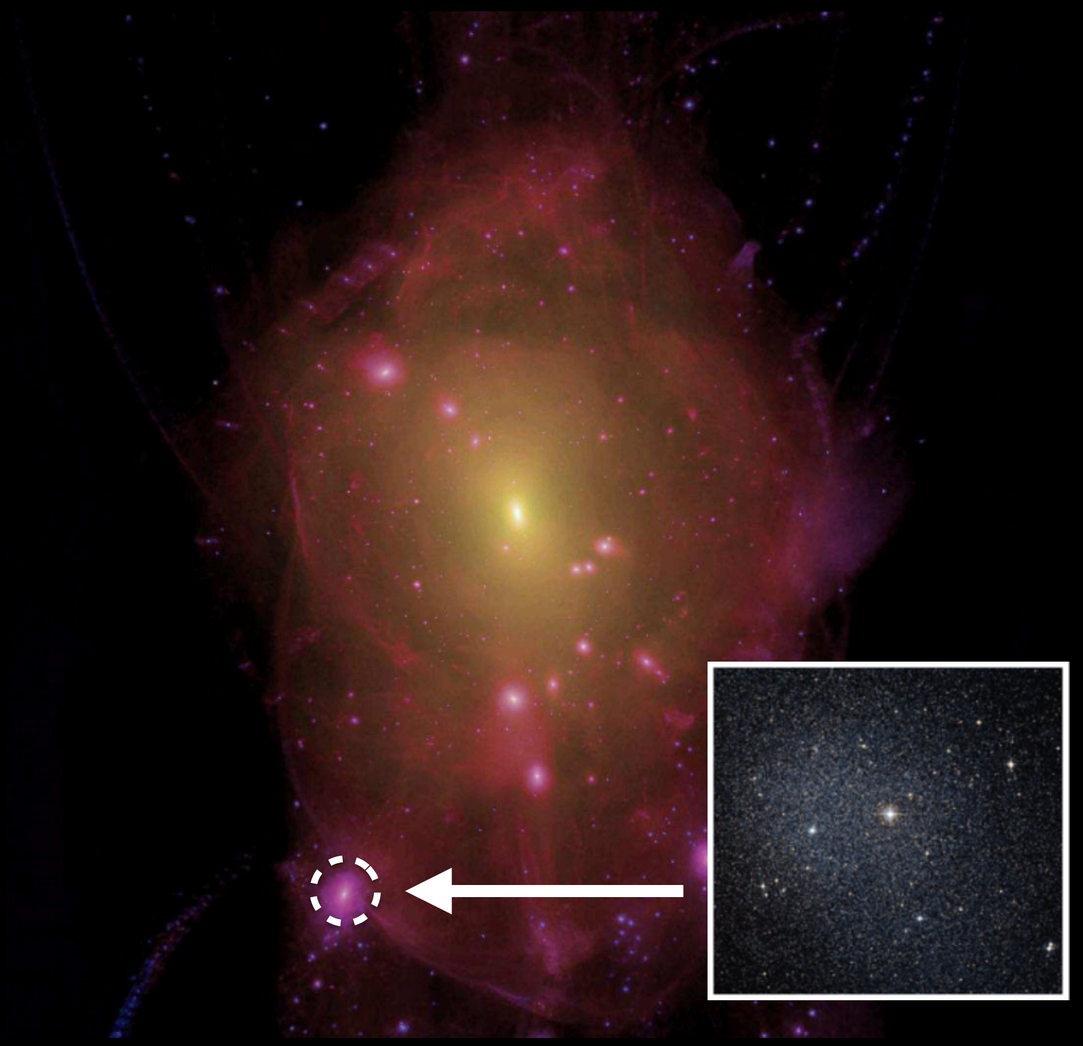
Galaxies Form in Dark Matter Halos

Dwarf Galaxies Test Dark Matter Models

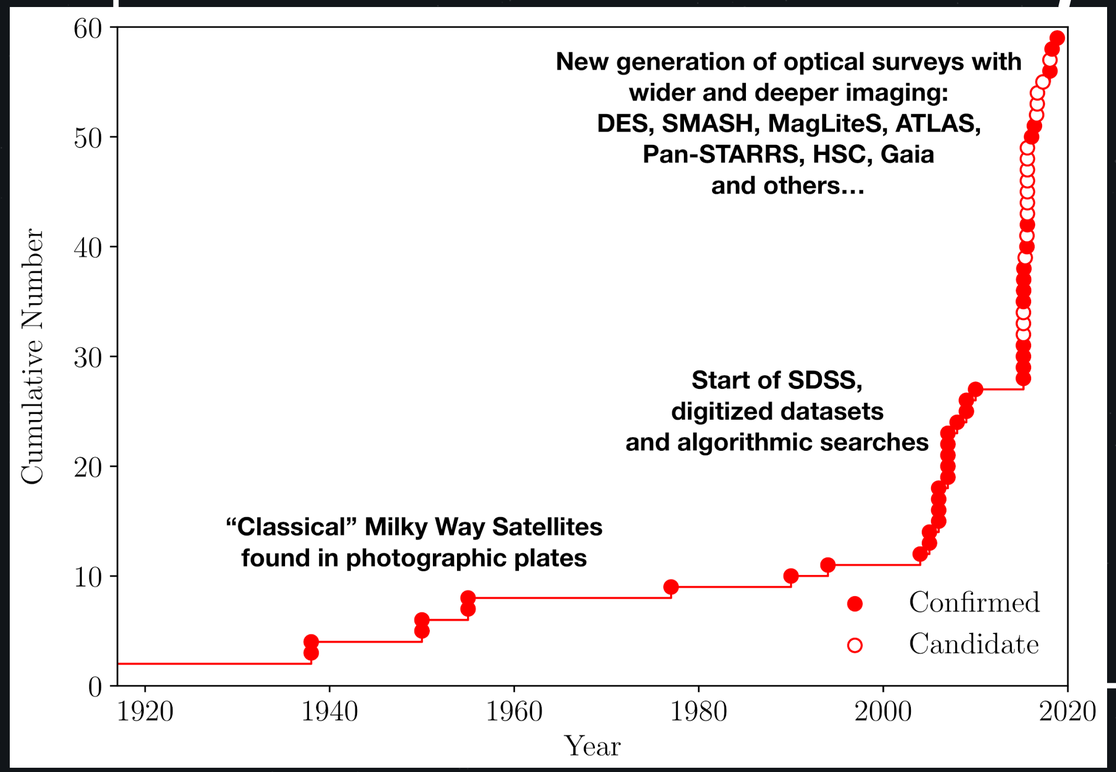
Cold Dark Matter

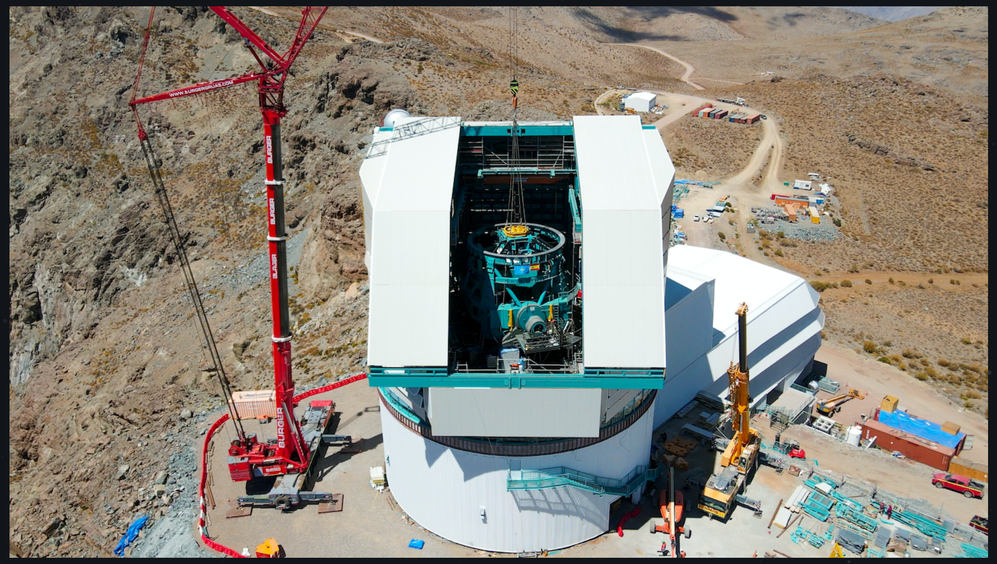


Warm Dark Matter



Upcoming Observations





Vera C. Rubin Observatory

Year

Discussion

Discussion

Discuss the following topics with your neighbor:

- What do you find most interesting/surprising/confusing from today's material?
- What questions do you have about the universe based on today's material?